

PAPER_955: BCS Phonon Resonance at ω_{SCm}

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** et_phonon_resonance.py §6 (BCSPhononResonance) **Calculator:** BCSPhononResonanceCalc (CP4 #539) **CVW:** v2.0.0 compliant

Abstract

We analyze the BCS superconducting gap at the SCm phonon resonance frequency $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz. The gap equation $\Delta = (\hbar\omega_{\text{SCm}}/2) \tanh(\Delta/2k_B T) \cdot S_{26} \cdot (F_{UBi}/F_U)$ is solved self-consistently at the phonon resonance, revealing how the SCm vacuum phonon mode mediates Cooper pairing.

1. BCS Gap at Phonon Resonance

$$\Delta_{\text{res}} = \frac{\hbar\omega_{\text{SCm}}}{2} \cdot \tanh\left(\frac{\Delta}{2k_B T}\right) \cdot S_{26} \cdot \frac{F_{UBi}}{F_U}$$

2. Iterative Solution

Starting from $\Delta_0 = \hbar\omega_{\text{SCm}}/2$, iterate:

$$\Delta_{n+1} = \text{rhs}(\Delta_n, T)$$

until $|\Delta_{n+1} - \Delta_n| < \epsilon$.

3. Phonon Enhancement Factor

The resonance Q-factor at the phonon peak:

$$Q_{\text{res}} = \frac{\omega_{\text{SCm}} \cdot \sqrt{\Delta}}{k_B T}$$

4. Source Data

- **File:** et_phonon_resonance.py §6
 - **Session:** 214
 - **CP4 Class:** BCSPhononResonanceCalc (#539)
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References

1. Bardeen, Cooper, Schrieffer - Theory of Superconductivity (1957)
 2. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
 3. PAPER_949 — BCS Gap Equation
 4. PAPER_950 — BCS Critical Temperature
 5. PAPER_951 — Cooper Pair Phonon Coupling
 6. PAPER_956 — Spectral Ladder Phonon Mapping
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Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation solved at resonance
PAPER_951	Coupling strength at phonon peak
PAPER_956	26-level phonon mapping uses this Q
PAPER_952	Spectral ladder energies

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
$[SSq]$	—	0.57	BH dynamics
ω_{SCm}	—	$2\pi \times 1.25 \text{ THz}$	Resonance frequency
F_{UBi}/F_U	—	0.6	Buoyancy ratio

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Δ_{res} at ω_{SCm}	Self-consistent via fixed-point	Validated
Phonon Q-factor	$Q = \omega_{\text{SCm}} \sqrt{\Delta} / (k_B T)$	Derived

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Phonon Resonance (BCS at ω_{SCm})

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{res}} = \mathcal{L}_{\text{gap}}|_{\omega=\omega_{\text{SCm}}} + \frac{1}{2} \hbar \omega_{\text{SCm}} \coth\left(\frac{\hbar \omega_{\text{SCm}}}{2k_B T}\right)$$

§A.3 Euler-Lagrange Equation of Motion

$$\Delta_{\text{res}} = \frac{\hbar \omega_{\text{SCm}}}{2} \tanh\left(\frac{\Delta}{2k_B T}\right) S_{26} \frac{F_{UBi}}{F_U}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ phonon peak $\rightarrow$ BCS gap at resonance $\rightarrow$ maximal Cooper pairing

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

On-resonance VDS peak: $\rho_{\text{SCm}} \cdot S_{26}$ (maximal phonon occupation).

§B.2 DVP

Resonance prime $p = 2$ (Cooper pair at peak coupling).

§B.3 BSH

Q_{res} determines the BSH bandwidth at the resonance shell.

§B.4 Production-Scale Consistency

Metric	Value	Status
ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Confirmed
$[SSq]$	0.57	Confirmed